



Stable isotopes reveal tissue-specific nutrient assimilation and energy allocation during ovarian maturation in the pink shrimp *Farfantepenaeus brasiliensis*

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Abstract

Stable isotope analysis provides an effective approach for investigating patterns of nutrient assimilation and resource use in aquatic organisms. Dietary assimilation in the pink shrimp *Farfantepenaeus brasiliensis* was assessed across four ovarian maturation stages using $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$. Muscle and ovarian tissues from 121 individuals were analysed to estimate the relative contributions of seven dietary source groups—algae, zooplankton, molluscs, crustaceans, fish, bivalves, and polychaetes—using the Bayesian mixing model MixSIAR. Algae represented the predominant dietary source in both tissues across all maturation stages, although their estimated contribution to muscle declined with advancing maturation. Zooplankton showed a probabilistic trend toward greater contribution to ovarian tissue during the developing and mature stages, whereas its contribution to muscle varied without a consistent directional pattern. Bivalves represented a secondary dietary source, with greater contributions to muscle during advanced maturation, while fish and polychaetes generally showed comparatively low contributions. Multivariate analysis revealed contrasting assimilation patterns between tissues, with muscle showing substantial overlap among maturation stages and ovarian tissue displaying greater stage-associated variation, particularly along the algae–zooplankton gradient. Overall, these results indicated tissue-specific differences in dietary assimilation associated with ovarian maturation. As maturation stages were unevenly represented across the sampling period, some seasonal influences may also be reflected in these stage-associated patterns. Differences between tissues may likewise reflect tissue-specific isotope turnover and temporal integration of dietary signals. These findings expand knowledge of trophic dynamics during reproduction in wild *F. brasiliensis* and provide a basis for future controlled studies on reproductive nutrition and broodstock diet formulation.

Keywords Dietary incorporation · Gonadal development · MixSIAR · Penaeid shrimp · Reproductive bioenergetics

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Introduction

Shrimp fisheries are predominantly based on species of the family Penaeidae (FAO 2024). These stocks remain under high exploitation pressure because of their global commercial importance (Diwan 2021; FAO 2024), with *Farfantepenaeus brasiliensis* (Latreille 1817) among the key target species in the southwestern Atlantic. Pink shrimp are highly valued in fisheries for their size and market demand, making them a valuable resource for both artisanal and industrial exploitation (da Rosa et al. 2021; FAO 2024). Penaeid shrimp fisheries are characterized by strong spatial structuring of life stages, with juveniles inhabiting estuarine environments and adults reproducing offshore, reflecting their estuarine–marine life-cycle connectivity (Castilho et al. 2015).

During reproduction, penaeid shrimp exhibit distinct gonadal development stages, ranging from immature to fully mature, with reproductive activity associated with advanced ovarian development (Craveiro et al. 2023). Ontogenetic dietary shifts are common in penaeids, and reproductive investment is closely linked to nutrient acquisition and allocation to gonadal tissues. In females, dietary inputs play a critical role in vitellogenesis and oocyte development, which depend on stored and assimilated energetic reserves (Diwan 2021; Craveiro et al. 2023).

In natural environments, adult penaeids are opportunistic benthic feeders, consuming a diverse range of prey items including polychaetes, bivalves, crustaceans, and detrital organic matter (Gutiérrez et al. 2016; Carvalho et al. 2021). Accordingly, recent studies have reinforced that penaeid shrimp exhibit high trophic plasticity, adjusting their diet according to local resource availability and environmental conditions (Carvalho et al. 2021, 2023). This opportunistic feeding behaviour justifies the inclusion of a wide range of potential sources in stable isotope mixing models to better capture the complexity of their diet in natural systems (Phillips et al. 2014). Finally, spatial and temporal variability in the availability and nutritional quality of these resources has been associated with differences in shrimp condition and reproductive output (Carvalho et al. 2023). Despite advances in penaeid biology, the physiological mechanisms governing nutrient allocation during reproduction remain partially elucidated, particularly regarding the partitioning of energy between somatic and reproductive tissues. Nutritional constraints remain a limiting factor for shrimp productivity in both wild populations and aquaculture systems, especially during gonadal maturation (Diwan 2021; Glencross 2021). However, the specific nutritional requirements of adult females during reproductive development remain incompletely understood (Glencross 2021). Broodstock nutrition represents a critical component of reproductive management in penaeid aquaculture because ovarian maturation requires substantial mobilisation and deposition of lipids, proteins, phospholipids, sterols, and essential fatty acids. Consequently, both nutrient composition and dietary ingredient quality can influence gonadal development, vitellogenesis, spawning performance, and egg quality. Recent feeding studies with *Penaeus vannamei* broodstock have demonstrated that phospholipid supplementation can enhance ovarian development and lipid deposition, while appropriate dietary cholesterol levels promote ovarian maturation, steroid synthesis, and reproductive performance (Liang et al. 2022, 2023; Ma et al. 2024). Likewise, krill-oil supplementation, as a source of phospholipids and long-chain fatty acids, has been associated with improved ovarian development and reproductive activity (Liang et al. 2024). These findings highlight the potential importance of matching broodstock diet composition to changing metabolic demands throughout ovarian maturation. Identifying which natural dietary pathways are increasingly represented in somatic and reproductive tissues

at different maturation stages may therefore provide an ecological framework for designing controlled studies on ingredient selection, nutrient balance, and stage-specific feeding strategies in penaeid broodstock. From a physiological ecology perspective, understanding how reproductive stage influences energy allocation is essential to elucidate trade-offs between somatic maintenance and reproduction in penaeid shrimp. Including multiple ovarian maturation stages establishes a clear biological framework, improving interpretation of shifts in energy use throughout reproduction (Peixoto et al. 2003; Tan-Fermin and Pudadera 1989). We therefore hypothesize that the proportion of assimilated dietary sources varies across stages of gonadal development and between muscle and ovarian tissues. Simultaneously evaluating muscle and ovary tissues offers a valuable tissue-specific approach, enabling a deeper understanding of differential nutrient allocation during the reproductive cycle, as increasingly recommended in recent ecological physiology studies (deVries et al. 2015). Stable isotope analysis of carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$) is widely used to investigate trophic relationships in aquatic ecosystems. This approach relies on predictable isotopic differences among primary producers and consumers, enabling the tracing of energy flow through food webs (Layman et al. 2012; Fry 2006). These isotopic differences arise from biochemical fractionation processes associated with assimilation and metabolic pathways. In shrimp ecology, stable isotopes have been extensively applied to identify dietary sources and trophic pathways in both natural and cultured systems (Layman et al. 2012). More recently, Bayesian mixing models have improved quantification of the proportional contributions of multiple sources, allowing explicit incorporation of uncertainty into dietary reconstructions (Stock and Semmens 2016; Parnell et al. 2013). Tissue choice may influence isotopic interpretation because metabolic activity and isotopic incorporation rates differ among tissues. Isotopic turnover is influenced by species-specific physiology, growth rates and tissue composition, leading to variation in temporal integration of dietary signals (Healy et al. 2018; Vander Zanden et al. 2015). Mixing models allow estimation of proportional dietary contributions, provided that appropriate trophic discrimination factors are applied (Parnell et al. 2013; Stock and Semmens 2016). Recent studies emphasize that uncertainty in trophic discrimination factors remains a key limitation in quantitative isotope ecology (Post 2002; Bond and Diamond 2011). Typical values for aquatic consumers generally range from ~ 0 to 1‰ for $\delta^{13}\text{C}$ and $\sim 2\text{--}4\text{‰}$ for $\delta^{15}\text{N}$, depending on diet composition and tissue type (Post 2002; McMahon and McCarthy 2016). The primary objective of this study is to evaluate dietary resource use in *Farfantepenaeus brasiliensis* across different reproductive stages using carbon and nitrogen stable isotope analysis, with emphasis on comparing the relative contributions of dietary sources among these stages. As a secondary objective, we assessed differences in nutrient assimilation and allocation between tissues by comparing isotopic contributions in muscle and ovary, providing insight into tissue-specific energy allocation during reproduction.

Material and methods

A total of 121 *F. brasiliensis* females were collected, comprising 31 immature (I-IMA), 34 rudimentary (II-RUD), 28 developing (III-DEV), and 28 mature (IV-MAT) individuals. The specimens were captured over the course of 1 year using a shrimp trawler in Ubatuba Bay ($23^{\circ}25'\text{--}23^{\circ}27'\text{S}$ and $45^{\circ}00'\text{--}45^{\circ}03'18''\text{W}$), São Paulo State, Brazil. Sampling was conducted offshore using multiple trawls at distinct fishing locations and during different months to minimize spatial and temporal clustering and reduce the risk of

pseudoreplication (Fig. 1). Individual shrimp were considered the biological unit of replication because gonadal maturation reflects an intrinsic physiological condition shaped by individual-specific factors such as age, genetics, and nutritional history. Sample sizes per gonadal-maturation stage and month ranged from 1 to 7 individuals, reflecting the natural occurrence and availability of each stage in the wild population. Individuals were neither pooled nor artificially balanced among maturation stages or sampling months.

Because this was an observational field study, the distribution of maturation stages across the sampling period was naturally unbalanced. Although each sampling event included different maturation stages when available, these stages were not equally represented throughout the year (Fig. 1). Consequently, maturation stage and sampling time were not fully independent factors in the sampling design. The multi-trawl, multi-location, and multi-month sampling strategy was therefore intended to provide replication across naturally occurring individuals while preserving the wild population's temporal and reproductive structure. For the application of the Bayesian mixing model, 36 potential food sources or prey items were selected—based on the known foraging behaviour of *F. brasiliensis*—and collected using a shrimp trawler from the same region where the shrimp were sampled. These food items were subsequently grouped into major categories (Table 1).

Stable isotope analyses of carbon and nitrogen were performed on the abdominal (tail) muscle and ovary tissues following standard protocols. Samples were thawed, oven-dried at 56 °C for 48 h, and homogenized by cryogenic grinding to obtain a fine and uniform powder.

Aliquots (300–350 µg) were weighed into tin capsules and analysed using an elemental analyser (Flash 2000, Thermo Scientific) coupled to an isotope ratio mass spectrometer (Delta V, Thermo Scientific). Combustion was carried out at 1020 °C, and resulting gases (CO₂ and N₂) were separated by gas chromatography prior to isotopic determination.

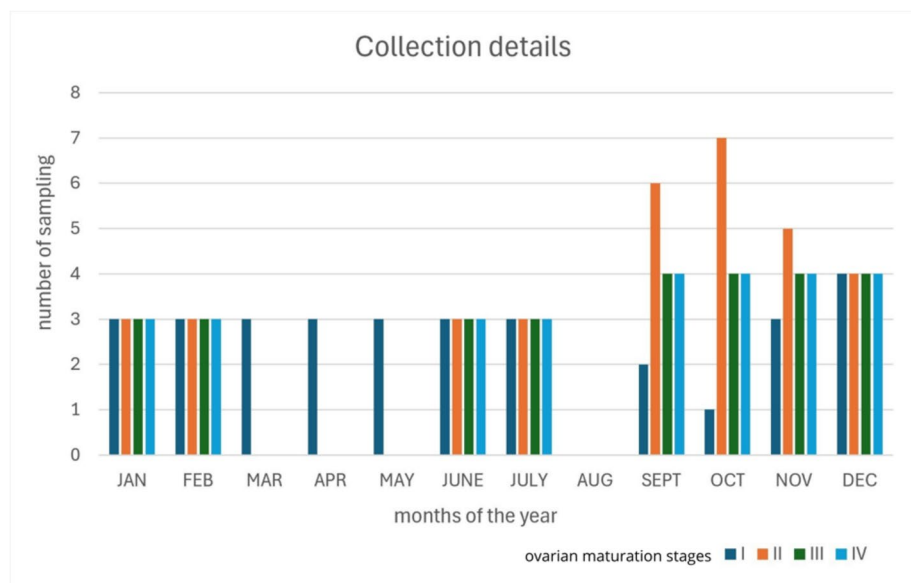


Fig. 1 Collection details. Legends: I, immature; II, rudimentary; III, developing; IV, mature. Note: In March, April, and May, the closed season occurs, during which only immature females (I) were captured under the authorization of a formal scientific collection permit (no. 72330). In August, no collection occurred due to adverse weather conditions

Table 1 Seven major groups of food resources with 36 species

Groups	Species	$\delta^{13}\text{C}$ (‰)	$\delta^{15}\text{N}$ (‰)
Algae	<i>Sargassum</i> sp.	− 19.43	+ 7.98
Zooplankton	<i>Chrysaora lactea</i>	− 20.77	+ 12.64
Molluscs	<i>Buccinanops cochlidium</i>	− 17.01	+ 12.77
	<i>Olivancillaria urceus</i>	− 17.33	+ 11.63
	<i>Fusinus verrucosus</i>	− 17.47	+ 11.06
	<i>Siratus tenuivaricosus</i>	− 17.18	+ 12.51
	<i>Olivella defiorei</i>	− 18.44	+ 11.67
	<i>Zidona dufresnei</i>	− 17.18	+ 13.20
	<i>Doryteuthis sanpaulensis</i>	− 19.83	+ 13.51
	<i>Loxopagurus loxochelis</i>	− 22.43	+ 7.51
Crustaceans	<i>Dardanus insignis</i>	− 18.77	+ 11.36
	<i>Pagurus exilis</i>	− 16.06	+ 10.00
	<i>Achelous spinimanus</i>	− 18.38	+ 12.69
	<i>Callinectes danae</i>	− 18.57	+ 11.84
	<i>Hepatus pudibundus</i>	− 17.76	+ 12.51
	<i>Persephona mediterranea</i>	− 14.10	+ 10.79
	<i>Stenocionops furcatus</i>	− 20.14	+ 12.05
	<i>Libinia spinosa</i>	− 20.09	+ 12.70
	<i>Chloroscombrus chrysurus</i>	− 18.80	+ 12.23
	<i>Stellifer rastrifer</i>	− 18.37	+ 13.33
	<i>Lagocephalus laevigatus</i>	− 17.71	+ 13.79
	<i>Micropogonias furnieri</i>	− 18.15	+ 13.47
	<i>Lutjanus synagris</i>	− 17.88	+ 14.74
	<i>Gymnothorax ocellatus</i>	− 18.46	+ 14.76
	<i>Bothus ocellatus</i>	− 19.61	+ 11.65
	<i>Chilomycterus antillarum</i>	− 18.92	+ 12.72
	<i>Paralanchurus brasiliensis</i>	− 19.05	+ 13.58
	<i>Ogcocephalus vespertilio</i>	− 19.29	+ 13.16
Fish	<i>Cathorops spixii</i>	− 18.33	+ 14.00
	<i>Upeneus parvus</i>	− 18.52	+ 11.78
	<i>Diplectrum radiale</i>	− 18.62	+ 11.42
	<i>Zapteryx brevirostris</i>	− 19.14	+ 13.19
	<i>Pseudobatos percellens</i>	− 18.82	+ 11.90
	<i>Atlantoraja castelnaui</i>	− 18.90	+ 13.51
Bivalve ^a	<i>Perna perna</i>	− 19.94	+ 11.08
Polychaete	<i>Spiochaetopterus nonatoi</i>	− 16.40	+ 10.02

^aSeparated from molluscs because bivalve is one of the most common ingredients used in maturation-inducing diet of penaeid broodstock (Moss and Crocos, 2001)

All procedures followed established methodologies for stable isotope analysis as described in the literature (Post 2002; Fry 2006), and methodological details have been streamlined in the revised manuscript to improve clarity and focus.

The $R(^{13}\text{C}/^{12}\text{C})$ and $R(^{15}\text{N}/^{14}\text{N})$ ratios in muscle and ovary samples were measured relative to international standards: the Vienna Pee Dee Belemnite (VPDB) for carbon and atmospheric nitrogen (AIR) for nitrogen. The ratio of stable isotopes, expressed in delta notation (δ), was calculated using the following equation:

$$\delta^i E = \frac{R(^i e/^j E)_{\text{sample}}}{R(^i e/^j E)_{\text{standard}}} \quad (1)$$

The symbology of the equation indicates that $R(^i E/^j E)$ represents the ratio between the less abundant and the more abundant isotopes, specifically $R(^{13}\text{C}/^{12}\text{C})$ and $R(^{15}\text{N}/^{14}\text{N})$. The δ value (sample vs standard) reflects the relative difference in isotope ratios of the chemical element in a given sample compared to its respective international standard, expressed in parts per thousand (‰). The isotope measurements were normalized using the certified reference materials USGS61, USGS62, and USGS63, following Paul et al. (2007) and Schimmelmann et al. (2016). The standard uncertainty of the analysis was estimated to be $\pm 0.10\text{‰}$ and $\pm 0.15\text{‰}$ for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, respectively.

Stable isotope mixing model

The selection of trophic discrimination factors (TDFs) of $+0.9\text{‰}$ for carbon ($\Delta^{13}\text{C}$) and $+2.5\text{‰}$ for nitrogen ($\Delta^{15}\text{N}$) was based on a comparative evaluation of multiple model scenarios, with the aim of accounting for variability in trophic fractionation and its implications for source contribution estimates. Variations in $\Delta^{13}\text{C}$ generally had limited influence on model outputs across all analysed groups, consistent with the relatively small carbon isotope fractionation between diet and consumer tissues (Fry 2006; McMahon and McCarthy 2016). By contrast, $\Delta^{15}\text{N}$ exerted a stronger influence on estimated source contributions, with responses differing by group. For algae (muscle and ovary), decreasing $\Delta^{15}\text{N}$ resulted in lower estimated contributions, whereas for crustaceans, molluscs, and fish, decreasing $\Delta^{15}\text{N}$ led to higher estimated contributions. This pattern reflects the known variability in nitrogen trophic enrichment associated with differences in diet, metabolism, and tissue-specific isotope incorporation (Post 2002; Bond and Diamond 2011; Vander Zanden et al. 2015). For mussels, polychaetes, and zooplankton, no consistent pattern was observed, indicating interspecific variability in trophic discrimination factors (Vanderkluft and Ponsard 2003; Bond and Diamond 2011). The use of an intermediate value of $+2.5\text{‰}$ for $\Delta^{15}\text{N}$ falls within the range commonly reported for aquatic consumers and helps reduce bias in source-contribution estimates when trophic enrichment is uncertain (Post 2002; McMahon and McCarthy 2016). Therefore, the combination of $+0.9\text{‰}$ ($\Delta^{13}\text{C}$) and $+2.5\text{‰}$ ($\Delta^{15}\text{N}$) was considered appropriate to represent trophic fractionation while maintaining comparability across trophic groups and tissues. To ensure the robustness of our mixing model outputs, a sensitivity analysis was conducted using alternative TDF scenarios for all seven dietary sources across both muscle and ovarian tissues. The results confirmed that the main stage-specific dietary trajectories were not artifacts of the chosen discrimination factors (detailed quantitative results are provided in Supplementary Table S1).

Multivariate analysis

To visualize and evaluate the integrated multidimensional shifts in dietary assimilation profiles across the four ovarian maturation stages, a principal component analysis (PCA) was

conducted. The PCA was performed using the posterior proportional estimates (dietary contributions) derived from the MixSIAR model for each tissue. Before the multivariate analysis, the compositional data were appropriately centred and scaled, and the principal components were computed using the *prcomp* function. PCA biplots were generated to map the spatial trajectories of somatic (muscle) and reproductive (ovarian) assimilation, with dietary sources plotted as vectors (contributions) to identify which resources drove differentiation between stages. To visually assess the degree of overlap and shifting patterns during maturation, 95% confidence ellipses were constructed around the multivariate space of each gonadal stage. All data wrangling, multivariate analyses, and graphical representations were performed within the R statistical environment (R Core Team 2023), utilizing the *tidyverse* package ecosystem for data manipulation, the *factoextra* package for PCA visualization and ellipse computation, and the *patchwork* package for combined graphical layout assembly.

Statistical analysis

Because this study utilized a Bayesian isotopic mixing framework (MixSIAR) to estimate dietary proportions, traditional frequentist statistical tests (e.g. ANOVA or Kruskal–Wallis) were not employed. Statistical inferences regarding differences in dietary assimilation across gonadal maturation stages and tissues were initially evaluated using the 95% Bayesian credible intervals (CrI) derived from the posterior distributions. Furthermore, given the inherent uncertainty and partial overlap of CrIs across stages, we additionally incorporated direct Bayesian pairwise comparisons. This approach calculated the exact posterior probability that the proportional contribution of a specific dietary source in one maturation stage was strictly greater than in another, providing a robust and direct probabilistic metric for evaluating directional dietary shifts.

Results

The measured isotopic means ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) and standard deviations for the *F. brasiliensis* muscle and ovarian tissues across all four maturation stages are detailed in Supplementary Table S2.

Dietary proportions and source contributions

Dietary assimilation profiles, estimated via the MixSIAR Bayesian mixing model, revealed that algae (AL) constituted the predominant basal resource assimilated into both muscle and ovarian tissues across all maturation stages. In muscle tissue, algal contribution peaked at 52.7% (95% CrI 35.2–72.8%) during the immature stage, then steadily decreased to 41.2% (95% CrI 24.2–63.4%) as maturation progressed (Table 2).

In the ovarian tissue, algal assimilation was high and remarkably stable, with posterior means ranging from 48.1% (95% CrI 25.9–74.7%) in mature females (IV-MAT) to 53.9% (95% CrI 35.9–73.5%) in immature individuals (I-IMA) (Table 3).

The most dynamic shift in dietary integration concerned zooplankton (PL) assimilation into ovarian tissue. While the 95% CrI showed considerable overlap, direct Bayesian comparisons indicated a high posterior probability that zooplankton contribution increased from the immature stage (Mean 7.9%; 95% CrI 0.3–22.8%) to the developing stage (Mean 27.4%; 95% CrI 1.7–56.8%), with an 86.7% probability that assimilation in stage III-DEV

Table 2 Proportional contribution (%) of food sources to the muscle tissue of *Farfantepenaeus brasiliensis*. Values are presented as mean (95% Bayesian credible intervals)

Food source	I-IMA	II-RUD	III-DEV	IV-MAT
Algae (AL)	52.7 (35.2–72.8)	46.6 (28.8–68.0)	41.9 (25.2–64.9)	41.2 (24.2–63.4)
Molluscs (MO)	7.3 (0.3–23.8)	7.2 (0.1–23.1)	7.6 (0.2–24.1)	7.9 (0.2–25.5)
Zooplankton (PL)	6.0 (0.2–18.1)	10.0 (0.4–27.6)	11.7 (0.5–30.4)	11.9 (0.5–30.8)
Bivalves (ME)	7.9 (0.2–26.9)	10.9 (0.3–34.8)	12.2 (0.3–37.5)	12.2 (0.3–39.5)
Crustaceans (CR)	9.7 (0.3–31.2)	10.3 (0.3–32.1)	11.1 (0.4–35.2)	11.1 (0.4–35.0)
Polychaetes (PO)	10.5 (0.5–29.0)	8.1 (0.3–22.7)	7.9 (0.3–23.2)	8.1 (0.3–22.9)
Fish (PE)	5.8 (0.2–19.8)	6.9 (0.1–23.1)	7.7 (0.2–25.1)	7.6 (0.2–25.2)

Table 3 Proportional contribution (%) of food sources to the ovarian tissue of *Farfantepenaeus brasiliensis*. Values are presented as mean (95% Bayesian credible intervals)

Food source	I-IMA	II-RUD	III-DEV	IV-MAT
Algae (AL)	53.9 (35.9–73.5)	52.7 (33.8–74.8)	49.9 (26.6–76.9)	48.1 (25.9–74.7)
Molluscs (MO)	6.4 (0.1–21.4)	5.4 (0.2–17.5)	3.6 (0.1–12.9)	4.3 (0.1–15.5)
Zooplankton (PL)	7.9 (0.3–22.8)	13.5 (0.6–34.9)	27.4 (1.7–56.8)	25.9 (1.6–53.8)
Bivalves (ME)	8.4 (0.3–27.8)	10.0 (0.3–31.6)	7.4 (0.2–24.4)	8.6 (0.2–27.6)
Crustaceans (CR)	9.4 (0.2–30.8)	8.1 (0.2–26.6)	4.8 (0.1–16.0)	5.8 (0.2–20.3)
Polychaetes (PO)	8.3 (0.4–24.0)	4.8 (0.1–15.2)	2.8 (0.1–9.4)	3.0 (0.1–10.1)
Fish (PE)	5.8 (0.2–18.6)	5.4 (0.2–18.6)	4.1 (0.1–14.7)	4.4 (0.1–15.3)

was greater than in stage I-IMA. A similar probabilistic trend was observed for mature stages (IV-MAT 25.9%; 95% CrI 1.6–53.8%), which showed an 85.7% probability of being greater than the immature baseline. Conversely, zooplankton assimilation into the muscle tissue remained relatively stable and lower across all gonadal stages, fluctuating narrowly between 6.0 and 11.9% (Table 2).

Bivalves (ME) consistently served as a secondary benthic energy source. Their assimilation was slightly higher in the muscle tissue (7.9 to 12.2%) compared to the ovaries (7.4 to 10.0%) throughout the reproductive cycle. Other benthic resources, such as crustaceans (CR) and molluscs (MO), displayed moderate background contributions with no pronounced stage-driven accumulation in either tissue. Polychaetes (PO), fish (PE), and molluscs (MO) generally showed among the lowest contributions in both tissues.

Multivariate analysis of dietary assimilation

To evaluate integrated multidimensional patterns of dietary assimilation across maturation stages and tissues, a principal component analysis (PCA) was performed on the posterior estimates (Fig. 2). For muscle tissue (Fig. 2A), the first two principal components explained 40.0% of the total variance (PC1 = 21.8%; PC2 = 18.2%). PC1 was primarily associated with algae (52.27%) and zooplankton (22.66%), which were oriented in opposing directions in the PCA ordination (Table 4). PC2 was predominantly associated with polychaetes (44.09%), followed by crustaceans (22.94%). The extensive

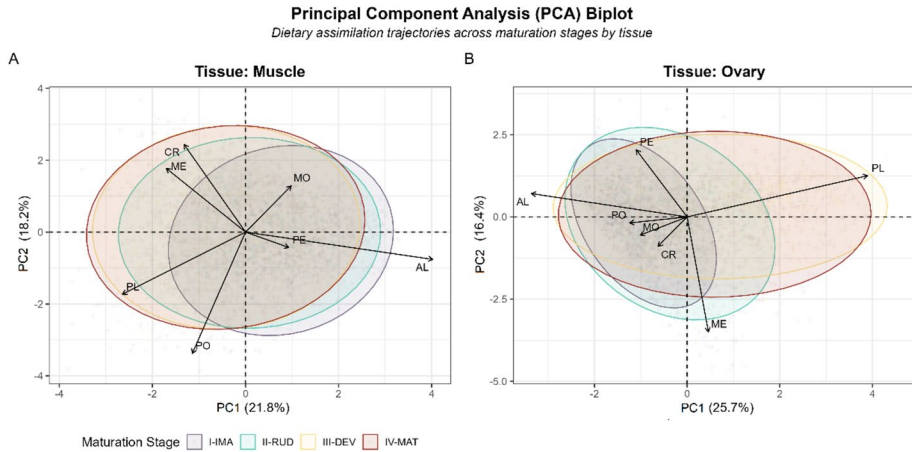


Fig. 2 A, B Principal component analysis (PCA)

overlap among the 95% confidence ellipses indicated limited multivariate differentiation in dietary assimilation among maturation stages in muscle tissue. In contrast, ovarian tissue exhibited greater multivariate displacement among maturation stages (Fig. 2B), with the first two principal components explaining 42.1% of the total variance ($PC1 = 25.7\%$; $PC2 = 16.4\%$). $PC1$ was primarily associated with zooplankton (49.14%) and algae (36.60%), which showed opposing vector orientations, whereas $PC2$ was predominantly associated with bivalves (62.41%), followed by fish (21.25%) (Table 4). Overall, the PCA indicated contrasting multivariate patterns between somatic

Table 4 Relative contribution (%) of the seven dietary source groups to the first two principal components ($PC1$ and $PC2$) for muscle and ovarian tissues in *Farfantepenaeus brasiliensis*

Food source	Contribution to $PC1$ (%)	Contribution to $PC2$ (%)
Muscle tissue		
Algae (AL)	52.27	2.22
Zooplankton (PL)	22.66	11.56
Bivalves (ME)	9.36	12.11
Crustaceans (CR)	5.62	22.94
Polychaetes (PO)	4.20	44.09
Molluscs (MO)	3.13	6.40
Fish (PE)	2.79	0.71
Ovary tissue		
Zooplankton (PL)	49.14	8.01
Algae (AL)	36.60	2.56
Polychaetes (PO)	5.02	0.17
Fish (PE)	3.92	21.25
Molluscs (MO)	3.31	1.54
Crustaceans (CR)	1.28	4.08
Bivalves (ME)	0.69	62.41

and reproductive tissues, with muscle showing substantial overlap among maturation stages and ovarian tissue displaying greater stage-related variation, particularly along the algae–zooplankton gradient.

Discussion

Considerable progress has been made in understanding nutrient acquisition and reproductive allocation in penaeid shrimp, particularly regarding the interaction between diet quality and gonadal maturation. By employing stable isotope analysis ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$), this approach provides an integrative and time-averaged perspective on nutrient assimilation, reducing biases commonly associated with instantaneous dietary assessments and offering a more reliable view of trophic dynamics (Quezada-Romegialli et al. 2018).

Current approaches increasingly emphasize physiological and ecological integration rather than purely empirical diet formulation, highlighting how reproductive stage is associated with variation in resource assimilation and tissue-specific allocation patterns (Craveiro et al. 2023).

Plant and filter-feeder sources (algae and bivalves)

Algae constituted the dominant basal resource in both muscle and ovarian tissues across all maturation stages. Their estimated contribution declined progressively across maturation stages in both tissues, with a more pronounced decrease in muscle than in the ovary. Although this pattern was associated with advancing gonadal development, maturation stages were not evenly distributed throughout the sampling year; therefore, temporally structured variation in algal availability and environmental conditions may also have contributed to the observed trend. Within this limitation, the contrasting trajectories between tissues are consistent with potential changes in tissue-specific resource use associated with reproductive development (Emerenciano et al. 2022; Carvalho et al. 2023).

Bivalves displayed complementary, tissue-specific trajectories: in muscle, their assimilation increased during the final maturation stages, coinciding with the decline in estimated algal contributions, supported by their richness in PUFAs, phospholipids, and sterols essential for structural maintenance (Liang et al. 2022, 2023). In ovarian tissue, bivalve contributions oscillated without a consistent directional trend, reflecting opportunistic assimilation modulated by prey availability rather than a linear reproductive demand.

The increasing contribution of bivalves to muscle tissue during advanced maturation, alongside their more variable incorporation into ovarian tissue, reinforces their established role as a nutritionally dense broodstock feed ingredient—rich in phospholipids, sterols, and structural lipids essential for membrane integrity and reproductive tissue maintenance (Liang et al. 2022, 2023). These results suggest that bivalves contribute differently to somatic vs reproductive compartments, which may matter when evaluating ingredient allocation in diets designed to support both body condition maintenance and gonadal development in captive broodstock.

Together, the combined reliance on algae and bivalves throughout maturation reflects a dual energetic strategy anchored in primary production and high-quality filter-feeder resources (Glencross 2021; Craveiro et al. 2023).

Animal benthic resources (crustaceans, molluscs, and polychaetes)

Animal benthic resources constitute an important component of the assimilated diet, particularly for somatic tissue. Crustaceans and polychaetes exhibit consistent incorporation patterns into muscle, indicating their role as reliable sources of protein and essential nutrients for maintenance and growth. These prey groups are widely recognised as key dietary components of penaeid shrimp due to their high digestibility and nutritional value, particularly in natural benthic environments (Craveiro et al. 2023).

Polychaetes contributed less to ovarian tissue during the developing and mature stages, whereas in muscle tissue, contributions oscillated without a consistent directional trend. This result is notable given the well-documented nutritional suitability of polychaetes for penaeid maturation under aquaculture conditions, attributed to their high levels of arachidonic acid (ARA), essential amino acids, and bioactive compounds that stimulate gonadal development (Craveiro et al. 2023). Their limited and declining isotopic representation in ovarian tissue under natural field conditions likely reflects low encounter rates or habitat-mediated prey selectivity constraints rather than nutritional inadequacy, highlighting an important distinction between the nutritional potential demonstrated under controlled aquaculture conditions and the effective dietary assimilation achievable in wild environments (Emerenciano et al. 2022; Carvalho et al. 2023).

The relatively modest and variable contribution of polychaetes to both tissues under natural conditions, despite their recognised importance as bioactive maturation-stimulating feeds in aquaculture (Anand et al. 2024), highlights a potential disconnect between the nutritional potential of a dietary ingredient and its actual assimilation in the wild. This distinction underscores the value of integrating stable isotope-based dietary assessments with controlled feeding trials to evaluate whether the assimilation patterns observed in nature are reproduced under captive feeding conditions.

A decline in mollusc contributions to ovarian tissue was observed during the final stages of gonadal maturation, contrasting with their increasing contribution to muscle along the ovarian development gradient. This opposing pattern indicates that molluscs were more closely associated with somatic support than with direct allocation to oocyte formation. Such tissue-specific routing of dietary resources is increasingly recognised as a key feature of crustacean reproductive bioenergetics, where the biochemical demands of each compartment differentially shape the assimilation and allocation of specific dietary inputs (Liang et al. 2023; Ma et al. 2024).

Fish and zooplankton (low- vs high-dynamic pathways)

Fish and zooplankton represent contrasting trophic pathways and differ markedly in their temporal dynamics and functional roles. Fish exhibit consistently low contributions across tissues and maturation stages, indicating a limited role in the assimilated diet. This pattern is consistent with the benthic feeding behaviour of penaeid shrimp, which typically exploit invertebrate prey rather than nektonic organisms (Carvalho et al. 2023).

In contrast, direct Bayesian comparisons suggest zooplankton serves as a dynamic resource, with a probabilistic trend toward greater incorporation into ovarian tissue during advanced maturation stages. This probabilistic trend should, however, be interpreted with the caveat that zooplankton abundance is itself markedly seasonal in coastal environments; its apparent association with advancing maturation stage may therefore partly reflect this temporal covariation rather than a purely physiological reproductive demand. Nonetheless,

this pattern remains consistent with the interpretation that zooplankton provides rapidly assimilated energy and nutrients preferentially allocated to reproductive processes, since fast-turnover prey such as zooplankton are known to play a key role in meeting periods of elevated energetic demand, particularly in marine invertebrates undergoing gonadal development (Griffen 2018).

Zooplankton, including *Artemia* biomass, is already recognised as an effective maturation feed component in penaeid broodstock diets due to its richness in long-chain polyunsaturated fatty acids (EPA and DHA), phospholipids, and sex steroids that stimulate gonadal development (Anand et al. 2024; Liang et al. 2022). The present results provide field-based support for the stage-specific relevance of this dietary category: rather than being uniformly important across maturation, the assimilation data suggest that zooplankton-derived nutrients become particularly critical from the mid-developing stage onward. This pattern could inform the timing of zooplankton or *Artemia*-enriched feed supplementation in captive maturation protocols, suggesting that increasing the proportion of this dietary component from the onset of the developing stage may better align with the physiological demands of ovarian tissue growth.

Aquaculture implications for broodstock nutrition and reproductive management

Taken together, these findings provide, for the first time in *F. brasiliensis*, an ecologically grounded reference for the relative importance of key dietary source groups across maturation stages and tissues under natural conditions. They generate testable, stage-specific hypotheses for controlled broodstock feeding experiments and support a more physiologically informed approach to maturation diet formulation—one in which ingredient selection and feeding regime are aligned with the stage-dependent nutritional demands of reproductive tissues rather than relying solely on empirically derived diet compositions.

Trophic integration, reproductive maturation, and broodstock nutrition

Overall, the combined evidence indicates that dietary assimilation in *F. brasiliensis* was structured around a persistent contribution of basal resources, complemented by benthic and pelagic pathways whose relative importance varied by maturation stage and tissue. Somatic tissues reflect a more conserved and integrative dietary signal, while reproductive tissues capture dynamic shifts associated with maturation.

These patterns reinforce the interpretation that penaeid shrimp exhibit flexible trophic strategies, integrating multiple energy pathways while modulating resource allocation between maintenance and reproduction. However, because maturation stages were not evenly distributed across sampling months, the observed isotopic patterns cannot be fully disentangled from seasonal and ontogenetic influences; they likely reflect the combined outcome of reproductive condition, environmental variability, and temporally structured resource availability, rather than isolated physiological effects. Accordingly, the present results describe dietary assimilation patterns associated with maturation stage in wild-caught females, but do not allow reproductive effects to be fully separated from this temporally structured ecological variability.

Additionally, consider differences in isotopic turnover between tissues when interpreting these patterns, as muscle integrates diet over longer periods, whereas ovarian tissue reflects more recent assimilation. This temporal decoupling may enhance the apparent contribution of dynamic resources in reproductive tissues (Phillips et al. 2014; Stock et al. 2018).

Isotopic values from different tissues may reflect dietary inputs across distinct temporal scales, which can influence interpretations of source contributions in mixing models. Therefore, interpret comparisons between tissues with caution, as differences in turnover rates may affect the temporal integration of isotopic signals (Phillips et al. 2014; Stock et al. 2018).

While this study investigates wild *F. brasiliensis*, the observed stage-specific assimilation trajectories provide valuable, testable hypotheses for optimizing broodstock management and diet formulation in aquaculture. The consistent reliance on algae and bivalves for baseline somatic maintenance suggests that maturation diets should maintain stable structural lipid and baseline protein inputs throughout the entire reproductive cycle. Conversely, the probabilistic trend toward increased zooplankton assimilation in ovarian tissue during advanced maturation indicates a stage-specific physiological demand for rapidly assimilated nutrients. In a captive setting, this implies that the timing of broodstock feeding could be refined by introducing highly digestible, high-turnover dietary ingredients, such as zooplankton-derived meals, specialized lipid emulsions, or targeted protein hydrolysates, specifically during the transition from rudimentary to developing ovarian stages to maximize reproductive performance.

However, it is important to consider the limitations of directly applying observations from wild populations to cultured broodstock. The dietary assimilation patterns observed here reflect natural environmental variability, opportunistic feeding behaviour, and seasonal changes in prey availability, which may differ from the more controlled conditions, stable environments, and formulated feeds used in aquaculture systems. Therefore, the stage-associated nutritional patterns identified in this study should be viewed as ecological reference points rather than as definitive dietary recommendations. Future controlled feeding trials will be important to evaluate these hypotheses and determine whether incorporating similar stage-specific nutritional profiles into formulated diets can support vitellogenesis, fecundity, and overall reproductive performance in captive penaeid broodstock.

Concluding remarks

Stable isotope analysis showed that *F. brasiliensis* consistently relied on algae as a primary energy baseline across all maturation stages and tissues. During maturation, changes in resource use indicated differential allocation between ovarian and muscle tissues rather than substantial shifts in overall diet. Zooplankton increased in importance for ovarian tissue, suggesting its role in reproductive investment, while benthic resources, particularly bivalves, contributed more to muscle, with their importance increasing along the maturation gradient. Fish played a negligible role throughout. Overall, the species exhibited a combination of trophic stability and flexibility, with evidence of differential nutrient routing between somatic and reproductive tissues during maturation.

This study provides, for the first time in *F. brasiliensis*, a high-resolution, tissue-specific profile of dietary assimilation across gonadal maturation stages under natural conditions. These findings expand the current understanding of reproductive bioenergetics in penaeid shrimp and provide a foundation for controlled feeding experiments to evaluate the nutritional requirements of broodstock.

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Data availability The data presented in this study are available in an word table format on request from the corresponding author.

Declarations

Ethics approval This research was conducted in accordance with the applicable Brazilian regulations for research involving free-living invertebrates. Biological material was collected under authorization No. 72330, issued by SISBIO/ICMBio/MMA in accordance with Normative Instruction No. 154/2007. The study was approved by the relevant research ethics procedures of São Paulo State University (UNESP).

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